A Hybrid Matrix Completion + Liquid Neural Network Framework for Groundwater Level Imputation

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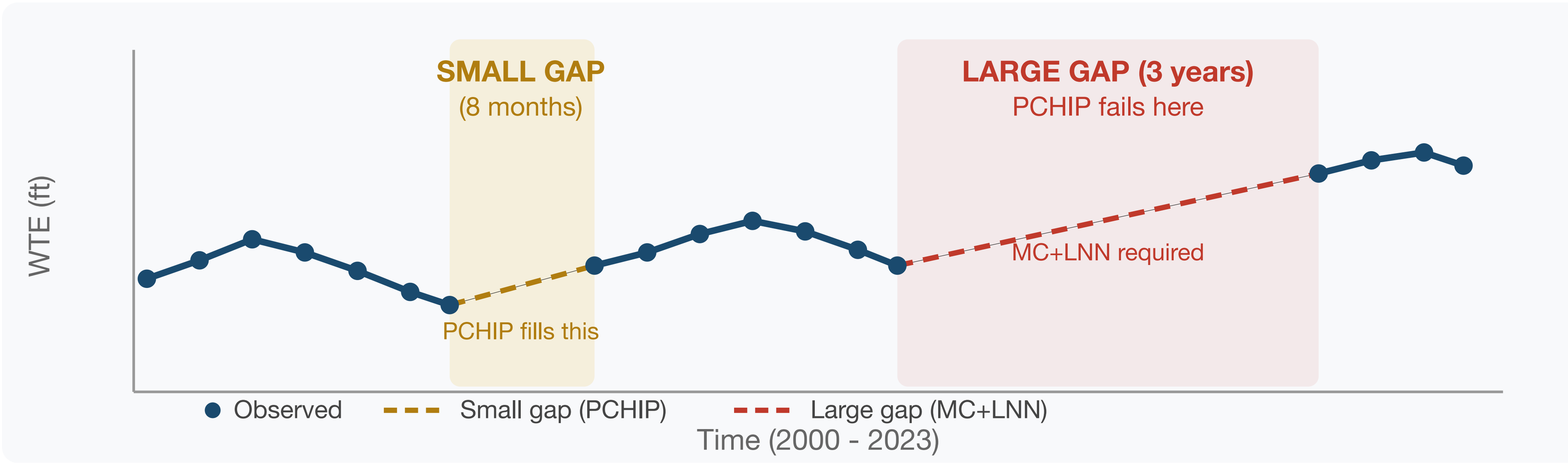
The Problem

Groundwater monitoring wells are the primary source of water-table elevation (WTE) data, yet public databases worldwide suffer from **long temporal gaps** -- periods of months to years with no measurements. Jasechko et al. (2024) analyzed 170,000 wells across 1,693 aquifer systems and found rapid declines in 30%, but most records are too incomplete for trend analysis.

This work introduces the first framework that jointly exploits spatial cross-well correlations and continuous-time temporal dynamics to reconstruct groundwater records with multi-year gaps.



What Is Matrix Completion?

Arrange all wells as columns and all months as rows in a matrix. Most entries are observed, but gap periods are missing. If wells share common patterns -- seasonal cycles, drought responses, recharge trends -- the matrix is approximately **low-rank**: a few shared patterns explain most of the variation.

Singular Value Decomposition (SVD) recovers these patterns. By iteratively decomposing the matrix, re-inserting known values, and repeating until convergence (**SoftImpute**), the missing entries are inferred from the structure of observed entries across all 592 wells simultaneously.

The composite matrix includes:

- **Target well row** -- observed values + gaps to fill
- **15 donor well rows** -- most correlated wells, weighted by Pearson r
- **5 GLDAS auxiliary rows** -- soil moisture at multiple time scales (globally available)
- **Seasonal encoding rows** -- sin/cos at 12-month period

The auxiliary and seasonal rows are **fully observed** for all time steps, anchoring the SVD and constraining predictions during gaps.

What Is a Liquid Neural Network?

A Liquid Neural Network (LNN) is a **continuous-time dynamical system** whose hidden state evolves according to an ordinary differential equation (Hasani et al., 2022). Unlike discrete-time networks (LSTM, GRU) that update at fixed steps, LNN's state flows continuously -- naturally accommodating the irregular sampling intervals of groundwater wells without resampling.

$$x(t + dt) = x \cdot e^{-\lambda \cdot dt} + (b / \lambda) \cdot (1 - e^{-\lambda \cdot dt}) \quad \text{where} \quad b = \tanh(W_{in} \cdot \text{input} + W_{res} \cdot x)$$

The **leak rate λ** controls how quickly the state forgets (fast leak = responsive to recent input; slow leak = long memory). The **input vector** concatenates the current observation (or MC placeholder during gaps), GLDAS soil moisture, and seasonal encoding. The **readout** is trained via ridge regression on observed values only -- ensuring fidelity to ground truth while using MC's spatial context to drive the reservoir through gap periods.

Hyperparameters (reservoir size 10-80 neurons, leak rate 0.05-0.95, input scaling 0.01-0.40) are auto-optimized via 8 trials per well using Kling-Gupta Efficiency. An ensemble of 3 models (different random seeds) selects the best.

The Combined Pipeline

Stage 1: PCHIP
Piecewise Cubic Hermite Interpolation fills **small gaps (≤24 months)** where bounding observations exist. Shape-preserving, no oscillation. Densifies the network so MC has better donor correlations.

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Stage 2: Matrix Completion
For **large gaps (>24 months)**: ARCHI selects top-15 correlated donors via OLS regression. SoftImpute fills the target row using the composite matrix (donors + GLDAS + seasonal). Adaptive SVD rank selection (3-12).

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Stage 3: LNN Refinement
LNN CFC refines MC predictions using **continuous-time dynamics**. MC output drives the reservoir as input, but readout trains only on real observations. Captures nonlinear temporal patterns MC's linear SVD cannot.

Key design principle: MC provides spatial context (what do nearby wells say?). LNN provides temporal dynamics (how does the system evolve between observations?). Neither alone achieves what the coupled framework delivers.

Applicability

The framework is **general-purpose**, not basin-specific. It requires only:

- **Well locations** (latitude, longitude) for donor correlation
- **Irregular observations** at any cadence (daily, monthly, quarterly)
- **GLDAS auxiliary data** -- globally available satellite-derived soil moisture (no local calibration needed)

All hyperparameters are auto-optimized per well. No manual tuning, no basin-specific training. Validated on the Great Salt Lake Basin (592 wells, 93,000 km², 24-year record) but designed for any monitoring network worldwide.

Cross-Validation Results

592 wells, Great Salt Lake Basin, 2000-2023. Most complete well used as target.

Random Missing Data (50 trials per level)

% Removed	KGE	R^2	RMSE (ft)
5%	0.837	0.771	2.53
20%	0.853	0.787	2.64
30%	0.844	0.770	2.79
50%	**0.847**	**0.788**	**2.74**

Consecutive Year Gaps (20 trials per level)

Gap Length	KGE	R^2	RMSE (ft)
1 year	0.783	0.703	2.98
3 years	0.802	0.730	3.02
5 years	**0.815**	**0.744**	**2.91**

0.85
KGE at 50% data removed

0.82
KGE at 5-year consecutive gap

Performance remains stable across all missing-data rates and gap lengths -- the spatial-temporal coupling provides consistent reconstruction regardless of how much data is lost.

Conclusions

- **First hybrid spatial-temporal imputation framework** for groundwater: matrix completion exploits cross-well correlations while liquid neural networks capture continuous-time dynamics conditioned on climate auxiliary data
- **KGE > 0.84** under random data loss up to 50%; **KGE > 0.78** for consecutive gaps up to 5 years -- substantially exceeding prior ELM-based methods that degrade below KGE 0.3 for gaps beyond 2 years
- **PCHIP small-gap filling** improves the full pipeline by 10-19% KGE by providing denser donor correlations for the MC stage
- **General-purpose:** Requires only well locations, irregular observations, and globally available GLDAS data. No basin-specific calibration. Applicable to any monitoring network worldwide

Key references: Hasani et al. (2022) *Nat. Mach. Intell.*; Candes & Recht (2009) *Found. Comput. Math.*; Evans et al. (2020) *Remote Sens.*; Ramirez et al. (2023) *Water*; Sharma et al. (2024) *Groundw. Sustain. Dev.*